

Nathan Gong

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EDUCATION

Columbia University | New York City, NY 2024 - 2026
Master of Science in Computer Science. GPA: 3.97/4.0
Coursework: High-Performance Machine Learning, Artificial Intelligence, Computer Vision, Computational Robotics

Northeastern University | Boston, MA 2019 - 2023
Bachelor of Science in Bioengineering. Minor in Computer Science. GPA: 3.96/4.0
Honors: University Honors Program, President's Award, Tau Beta Pi, PEAK Research Award

SKILLS

Languages: Python, C++, Java, JavaScript, SQL, C, Bash
Libraries/Frameworks: PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib, Flask, Spring Boot, Pytest, GoogleTest
Tools/Platforms: Git, Docker, GCP, Jira, Bamboo, Linux, QNX, Windows, MySQL, MongoDB, Neo4j, Jupyter
Practices: TDD, OOP, Agile, CI/CD, SDLC

EXPERIENCE

Software Engineer II | *Medtronic* | Boston, MA Jan 2022 - Present

- Develop production software in an embedded, mixed-OS (Linux, QNX, Windows) environment for a surgical robotics platform, the company's highest-funded product
- Spearheaded failsafe system integration to reduce error recovery time by 70% and enhance reliability during clinical procedures, impacting hundreds of cross-functional stakeholders and accelerating U.S. regulatory clearance
- Refactored the core GUI framework onto the new failsafe architecture and built an alarm routing tool to propagate messages across distributed compute nodes, enabling end-to-end testability and improving real-time error visibility
- Implemented core C++ library enhancements for the platform migration to a unified OS, hardening system security through least-privilege execution of services and application image authentication
- Designed and validated manufacturing tooling supporting a production line transfer, and strengthened build scripts with structured error handling and CI-integrated unit test suites
- Built a full-stack Python application to aggregate web data and generate risk reports for 30+ off-the-shelf software products, streamlining the user experience and improving efficiency of analyses by 80%

Teaching Assistant | *Northeastern University* | Boston, MA Sep 2020 - Aug 2022

- Mentored 100+ students by providing technical guidance, debugging code, and explaining core algorithmic concepts

Software Engineer Co-op | *iRhythm Technologies* | San Francisco, CA Jan 2021 - Jun 2021

- Automated firmware testing infrastructure for an ambulatory cardiac monitoring device to reduce test time by 50%
- Designed APIs to interface with manufacturing software tools and configure embedded systems for production

PROJECTS

Roofline-Aware LLM Inference Analysis Jan 2026 - May 2026

- Profiled Llama-3.1-8B inference on an NVIDIA L4 GPU with FlashAttention-2 and 4-bit quantization to quantify a 63% memory reduction at an 18% decode-throughput cost

Robotic Tree Pruning Simulation Sep 2025 - Dec 2025

- Built a motion-planning pipeline to autonomously prune a simulated tree with a robotic arm - segmenting 3D point clouds to localize cut points, reconstructing surface meshes for collision modeling, and planning cutting trajectories

Movie Data Engineering Tool Nov 2024 - Dec 2024

- Engineered an ETL pipeline and web application to scrape and clean thousands of records from online sources, persist them across relational/document/graph stores, and visualize movie and actor popularity trends